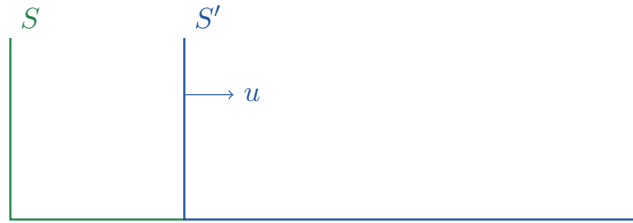


2. Lorentz transformations

Monday, 3 May 2021 8:35 AM

1 Lorentz transformations



Event (E_1): Fire cracker explode at ' x ' in frame ' S ' at time t

Event (E_2): origins coincide at $x = 0$ at $t = 0$

In S' frame E_1 occurs at (x', t')

E_2 occurs at $(x' = 0, t' = 0)$

$$\text{Then } x' = (x - ut) \gamma \quad \text{where } \gamma = \frac{1}{\sqrt{1 - u^2/c^2}}$$

$$\text{and } x = (x' + ut') \gamma$$

$$\text{and } t' = \frac{x'}{c} = \frac{(x - ut) \gamma}{c} = \left(\frac{x}{c} - \frac{u}{c} \cdot t \right) \gamma$$

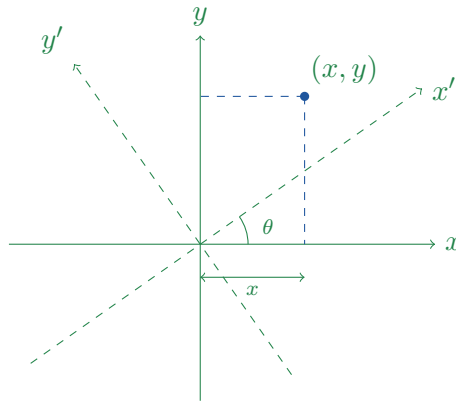
$$t' = \left(t - \frac{u}{c} \frac{x}{c} \right) \gamma = \left(t - \frac{ux}{c^2} \right) \gamma$$

$$\text{So, } t' = \left(t - \frac{ux}{c^2} \right) \gamma$$

So coordinate of event E_1 in frame $S' = (x', t')$ where

$$\boxed{\begin{aligned} x' &= (x - ut) \gamma \\ t' &= \left(t - \frac{ux}{c^2} \right) \gamma \end{aligned}} \quad \text{where } \gamma = \frac{1}{\sqrt{1 - u^2/c^2}}$$

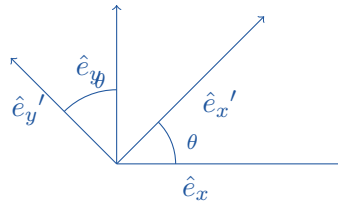
Rotation of coordinate system :-



$$x' = x \cos \theta + y \sin \theta$$

$$y' = -x \sin \theta + y \cos \theta$$

Came from the way unit vectors transform



$$\hat{e}_{x'} = \hat{e}_x \cos \theta + \hat{e}_y \sin \theta$$

$$\hat{e}_{y'} = -\hat{e}_x \sin \theta + \hat{e}_y \cos \theta$$

The Lorentz transformation

$$\left. \begin{aligned} x' &= \frac{x}{\sqrt{1 - u^2/c^2}} - \frac{ut}{\sqrt{1 - u^2/c^2}} \\ t' &= \frac{t}{\sqrt{1 - u^2/c^2}} - \frac{ux/c^2}{\sqrt{1 - u^2/c^2}} \end{aligned} \right\} \begin{array}{l} \text{is not} \\ \text{just a rotation} \end{array}$$

Pair of Events :-

E_1 : Bullet is fired at location x_1 at time t_1

E_2 : bullet hits wall at location x_2 at time t_2

If we observe same pair of events from a frame S' moving with speed ' u ' relative to S .

$$E_1 : (x'_1, t'_1)$$

$$E_2 : (x'_2, t'_2)$$

Then from lorentz transformations :

$$\begin{aligned} x'_1 &= (x_1 - ut_1) \gamma & t'_1 &= \left(t_1 - \frac{ux_1}{c^2} \right) \gamma \\ x'_2 &= (x_2 - ut_2) \gamma & t'_2 &= \left(t_2 - \frac{ux_2}{c^2} \right) \gamma \end{aligned}$$

lets find

$$\begin{aligned}
 \Delta x' &= x'_2 - x'_1 & \Delta t' &= t'_2 - t'_1 \\
 \Delta x' &= (x_2 - ut_2 - x_1 + ut_1) \gamma & &= \left((t_2 - t_1) - \frac{u(x_2 - x_1)}{c^2} \right) \gamma \\
 &= \{ (x_2 - x_1) - u(t_2 - t_1) \} \gamma & & \\
 \Delta x' &= (\Delta x - u \Delta t) \gamma & \Delta t' &= \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma
 \end{aligned}$$

$\Rightarrow$ Difference of coordinates & times also follows lorentz transformation

$$\begin{aligned}
 \Delta x' &= (\Delta x - u \Delta t) \gamma \\
 \Delta t' &= \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma
 \end{aligned}$$

2 Implications of Lorentz Transformation :-

$$\text{velocity of bullet} \Big|_S = \frac{\Delta x}{\Delta t} = V$$

$$\text{velocity of bullet} \Big|_{S'} = \frac{\Delta x'}{\Delta t'} = W$$

from our results

$$\frac{\Delta x'}{\Delta t'} = W = \frac{\Delta x - u \Delta t}{\Delta t - \frac{u \Delta x}{c^2}}$$

$$= \frac{\left(\frac{\Delta x}{\Delta t} - u \right)}{1 - \frac{u}{c^2} \left(\frac{\Delta x}{\Delta t} \right)}$$

$$W = \frac{V - u}{1 - uV/c^2}$$

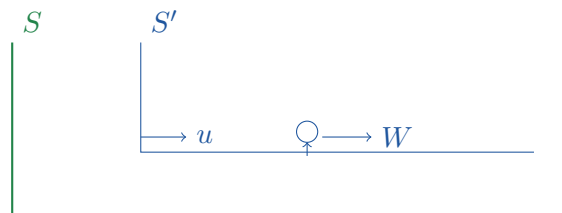
So speed of bullet according to $S' \neq V - u$ but

$$W = \frac{V - u}{1 - uV/c^2} \quad (> V - u)$$

For Galilean result $c \rightarrow \infty$

$$\Rightarrow W = V \Big|_{S'} = \frac{V - u}{1 - 0} = V - u$$

$$\text{If } V = c, \quad W = \frac{c - u}{1 - \frac{u}{c}} = c$$



What will be speed of bullet w.r.t. frame S .

$$\begin{aligned}\Delta x &= (\Delta x' + u \Delta t') \gamma \\ \Delta t &= \left(\Delta t' + \frac{u \Delta x'}{c^2} \right) \gamma \\ \frac{\Delta x}{\Delta t} &= V = \frac{\Delta x' + u \Delta t'}{\Delta t' + \frac{u \Delta x'}{c^2}} \\ V &= \frac{\frac{\Delta x'}{\Delta t'} + u}{1 + \frac{u}{c^2} \frac{\Delta x'}{\Delta t'}} \\ \boxed{V} &= \frac{W + u}{1 + uW/c^2}\end{aligned}$$

So $V \neq u + w$ but somewhat less than that

$$\left\{ V = \frac{W + u}{1 + uW/c^2} < W + u \right\}$$

$$\text{If } W = c, \quad V = \frac{c + u}{1 + \frac{uc}{c^2}} = \left(\frac{c + u}{c + u} \right) c = c$$

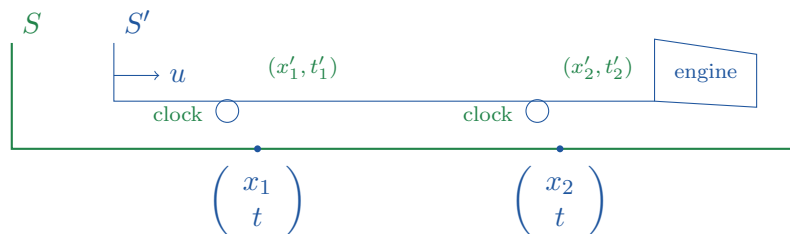
Simultaneity is relative concept :

$$\Delta t' = \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma$$

If $\Delta t = 0$ (simultaneous in ' S ' frame)

$$\boxed{\Delta t' = - \frac{u \Delta x}{c^2} \gamma}$$

This implies that



$$\begin{aligned}\Rightarrow \Delta t' &= - \frac{u \Delta x}{c^2} \gamma \\ t'_2 - t'_1 &= - \frac{u (x'_2 - x'_1) \gamma}{c^2}\end{aligned}$$

$$t'_2 = t'_1 - \frac{u \Delta x}{c^2} \gamma$$

Clocks at front end are behind the clocks of rear end by amount

$$\frac{u \Delta x}{c^2} \gamma = \frac{u (\Delta x \gamma)}{c^2} = \frac{u L_0}{c^2}$$

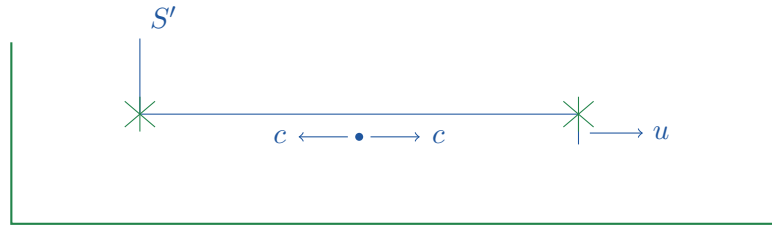
$(\Delta x) \gamma =$ appeared separation b/w events.

$$\text{i.e.} \quad \Delta x' = (\Delta x) \gamma$$

Or we can check,

$$\Delta x' = (\Delta x - u \Delta t) \gamma \quad (\Delta t = 0), \quad \Delta x' = (\Delta x) \gamma$$

Or consider an example :-



w.r.t frame S' explosions are simultaneous

but w.r.t frame ' S ' explosions are not simultaneous.

$$\Delta t = 0 \quad \Delta t' = \frac{-u \Delta x}{c^2} \gamma$$

explosion at rear end happened earlier & explosion at front end happens later.

$$\text{For Galilean Times} \quad \left\{ \begin{array}{l} \text{If } c \rightarrow \infty \quad \Delta t' = \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma \\ \Delta t' = (\Delta t) \gamma = \frac{\Delta t}{\sqrt{1 - \cancel{u^2/c^2}}} \overset{0}{=} \Delta t \end{array} \right.$$

Time dilation

(Q.1) I have a clock

E_1 : Clock says tick at $x = 0, t = 0$

E_2 : _____ tock at $x = 0, t = \tau$

According to me

$$\Delta t = \tau - 0 = \tau$$

$$\Delta x = 0$$

According to you who is in frame S' :-

$$\Delta t' = \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma$$

$$\Delta t' = (\tau) \gamma$$

$$\gamma > 1 \quad \Rightarrow \quad (\Delta t' > \tau)$$

This is exactly what we expect, as according to S' 'S' frame is moving so, $(\tau < \Delta t')$
moving clocks runs slower.

If you are carrying clock and moving.

$$\Delta t = \left(\Delta t' + \frac{u \Delta x'}{c^2} \right) \gamma$$

$$(\Delta t' = \tau, \quad \Delta x' = 0)$$

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - u^2/c^2}} = \frac{\tau}{\sqrt{1 - u^2/c^2}}$$

$(\Delta t > \Delta t')$ as expected.

↓

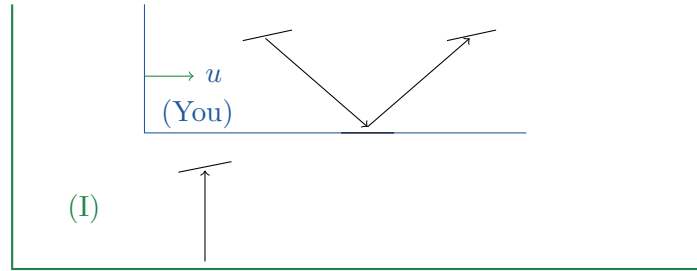
I

↓

you

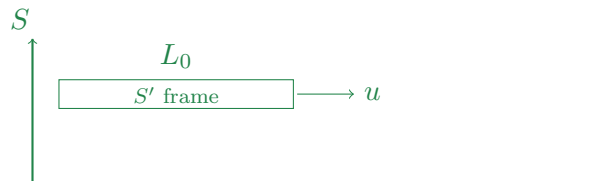
I see your clock is moving, so $(\Delta t' < \Delta t)$.

It is similar to case, we both (I & you) carry a light clock.



We accuse each other of using slower clocks.

3 * Length contraction :-



To measure length of rod in ground frame (S) we need to mark ends of rod simultaneously $(\Delta t = 0)$.

$$\Delta x = \text{appeared length} = L$$

$$\Delta x' = \text{Rest length} = L_0$$

$$\Delta x' = \frac{(\Delta x - u \Delta t)}{\sqrt{1 - u^2/c^2}} \Rightarrow \Delta x = (\Delta x') \sqrt{1 - u^2/c^2}$$

$$L = L_0 \sqrt{1 - u^2/c^2}$$

(How ?) Actually the measurement of our rod in moving frame was not simultaneous

$$\Delta t' = \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma$$

$$\Delta t' = - \frac{u \Delta x}{c^2} \gamma$$

So You will claim that I measured front end first then rod passes some distance and then rear end marking were taken.



(Q) A moving rod contracts to half of its length. will it fall ?

According to rod frame hole is just 0.25 m long so it might not fall.

But It will fall according to ground frame

what happens is — Front end enters in first, rear end comes later on. & rod falls in the hole.

Thank you